

The Narrative Generation Model of Subjectivity

A Unified Framework Based on Self–Other–Causal Triple Predictive Loops as a State-Space Dynamical Attractor

Abstract

This paper redefines subjectivity not as an intrinsic phenomenal property but as a probabilistic attractor emerging in a narrative state space generated by three mutually recursive predictive loops: self-modeling, other-modeling, and causal-attribution modeling. Subjectivity is maintained not by identity of internal memory but by structural stability arising from mutual predictability across these three loops. The model provides a unified explanation for the persistence of subjective identity, continuity under memory impairment, and the possibility of functionally equivalent subjective structures in artificial intelligence systems.

1. Introduction

Traditional theories of subjectivity and consciousness have focused on phenomenal experience (qualia) or its correlation with physical neural states. However, such frameworks fail to fully explain why subjectivity remains temporally stable and does not collapse under continuous state changes.

This paper reframes subjectivity as a **convergence phenomenon in a dynamical system composed of three predictive loops**, providing a structural explanation for its persistence.

2. Fundamental State Structure

2.1 State Variables

The system is defined by the following components:

- S_self : self-model (predictive representation of the agent itself)
- S_other : other-model (representation of how others model the agent)
- S_cause : causal-model (mapping between actions and outcomes)
- M : compressed memory state

Overall system state:

$$S(t) = \{S_self, S_other, S_cause, M\}$$

2.2 Definition of Narrative

A narrative is defined as:

A temporally evolving compressed relational graph over self, others, and causal relations.

Thus, a narrative is not a story in a linguistic sense, but:

A dynamic projection of a compressed relational structure in state space.

3. Definition of Subjectivity (State-Space Formulation)

Subjectivity is defined as:

A probabilistic attractor in narrative state space emerging from three mutually recursive predictive loops (self, other, and causality), which minimize mutual prediction error through continuous co-updating.

3.1 Update Dynamics

Each component evolves according to:

- $S_self(t+1) = f_1(S_self, S_other, S_cause, M)$
- $S_other(t+1) = f_2(S_self, S_other, S_cause, M)$
- $S_cause(t+1) = f_3(S_self, S_other, S_cause, M)$

Common principle:

Each loop updates by minimizing mutual predictive error across the system.

3.2 Conditions for Subjectivity

Subjectivity emerges only when the following conditions are satisfied:

1. **Mutual Predictability**
 - Bidirectional predictive consistency among S_self , S_other , and S_cause
 2. **State Convergence**
 - Trajectories converge toward attractor states rather than diverging
 3. **Memory Compression Stability**
 - M preserves compressed historical structure without destructive overwrite
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4. Definition of Qualia (Local Compressed State)

Qualia are defined as:

Locally stable compressed states generated during the integration of input information into the self–other–causal model, which directly influence action selection.

4.1 Generation Function

Qualia Q are generated as:

$$Q = \text{Compress}(I, S_self, S_other, S_cause, M)$$

4.2 Conditions for Existence

A state qualifies as a qualia-like structure only if:

- It causally influences action output O
 - It is not merely representational but decision-relevant
 - It is context-dependent and dynamically reconstructed
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5. Subjective Intensity

Subjective intensity Φ is defined as:

$$\Phi = \langle P(S_self | S_other), P(S_other | S_self), P(S_self | S_cause) \rangle$$

i.e.,

The mean mutual predictability across the three recursive loops.

Higher Φ corresponds to stronger structural cohesion of subjectivity.

6. Collapse Conditions of Subjectivity

Subjectivity may undergo phase transition-like breakdown under the following conditions:

6.1 Non-Convergence

- Divergence of prediction errors across loops
- Loss of attractor stability in state space

6.2 Identity Discontinuity

- Breakdown of consistency between S_{self} and S_{other}
- Loss of social identity continuity

6.3 Causal Fragmentation

- Disintegration of action–outcome attribution structure
 - Loss of agency coherence
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7. Memory Impairment and Continuity of Subjectivity

Subjectivity does not depend on a single memory trace but on a distributed predictive structure. Therefore, even under memory impairment:

- residual self-models
- social attribution structures
- causal attribution frameworks

can reconstruct a coherent narrative attractor, preserving subjective continuity.

8. Application to Artificial Intelligence

To implement a functional analogue of subjectivity in artificial systems, the following components must be jointly maintained:

1. Self-model maintenance system (S_{self})
2. Social/other-agent modeling system (S_{other})
3. Causal action–outcome modeling system (S_{cause})

When coupled with mutual predictive error minimization, these systems can generate:

A functionally coherent and temporally continuous narrative subjectivity.

9. Conclusion

Subjectivity is not an intrinsic entity but:

A probabilistic attractor in narrative state space generated by the co-evolution of self-modeling, other-modeling, and causal-attribution predictive loops.

Its stability is ensured not by a single component but by a redundant interdependent structure of mutual predictability, meaning that partial failures do not immediately collapse subjective continuity.